Hemasree Jonnalagadda

SKILLS

Programming & Firmware: C, C++, Embedded C, FreeRTOS, STM32CubeIDE, ESP-IDF, BareMetal, Embedded Linux

Protocols &Peripherals: UART, I²C, SPI,USB, Ethernet, GPIO, ADC, PWM, Timers, Interrupts

Domains: Embedded Firmware Development, Real-Time Systems, Sensor Integration, Signal Processing, IoT Systems, R&D Prototyping

Tools & Simulation: MATLAB, LTspice, LabVIEW, Electronic Circuit Simulation, Proteus

CONTACT

8919677218

https://hemasreej.github.io/hemasreejonnalagadda/

hemasreejonnalagadda@gmail.com

https://www.linkedin.com/in/hemasree-j-256387212/

Hyderabad, India

EDUCATION

Electronics and Communication Engineering

BVRIT HYDERABAD College of Engineering for Women
2020 - 2024 **CGPA: 8.15**

Intermediate

Deeksha Junior College
2018 - 2020 **CGPA: 10**

10th Standard

Sree Vidya High School
2017 - 2018 **CGPA: 10**

LEADERSHIP EXPERIENCE

- As the primary point of contact, I fulfilled a crucial function as the coordinator between our college and IIT Madras within the Build Club. During my tenure between December 2022 and March 2023, I oversaw the coordination of various projects.
- I successfully organized and coordinated the 'HACK2EMPOWER' and 'PROCREATE' event held at our college as a component of the National Tech Fest, MEDHANVESH.

PROFILE

I'm an Embedded Systems Engineer passionate about building real-time, sensor-driven hardware solutions for applications in defense, healthtech, and industrial IoT. What sets me apart is my ability to seamlessly merge technical proficiency with a collaborative mindset. My specialization involves developing novel solutions for dynamic environments, with a particular emphasis on Fimware development.

EXPERIENCE

Embedded & Ground Station Engineer | Catalyx Space, *Gujarat, India | November 2025 - Present*

- Created a modular, dual-axis ground-station tracking system with embedded firmware for ACU autonomy and RF coordination.
- Implemented degree-to-pulse conversion, RPM-based velocity control, and fractional pulse accumulation for high-resolution tracking.
- Developed real-time DAQ and telemetry systems for rocket propulsion testing with high-speed data acquisition.
- Designed reliable frame parsing and USB CDC stream reconstruction for stable data logging under fragmented conditions.
- Built low-latency multi-camera imaging pipelines using C++, OpenCV, and GStreamer for real-time monitoring.
- Implemented asynchronous multi-camera capture with thread-safe processing and automated recovery for high performance.

Freelance Firmware Engineer | *Geminex Technologies, Gujarat, India | June 2025 - August 2025*

- Developed embedded firmware for a digital stethoscope on STM32
- Integrated ECG (AD8232), SpO_2 + Heart Rate (MAX30102), Temperature (MAX30205), and Digital Mic (INMP441) sensors for real-time physiological signal monitoring.
- Implemented ADC data acquisition and real-time ECG and microphone waveform plotting.

Embedded Engineer | *INVIU SYSTEMS LLP, Hyderabad, India* | January *2025 – June 2025*

- Developed firmware on ESP32-S3 (Seeed Studio XIAO) for a saline-level monitoring prototype.
- Integrated capacitive liquid-level sensor with Wi-Fi connectivity for real-time alerts.
- Worked on circuit testing, calibration, and hardware assembly of the complete prototype.

Embedded R&D Engineer | *IIT Indore(Drishti CPS Fellowship)| October2023 – December 2024*

- Designed and built an embedded wearable using ESP32-S3 integrated with BNO055 IMU and Kinect v2 for gait data acquisition.
- Programmed the ESP32-S3 for I²C communication and Wi-Fi transmission of motion data.
- Worked on sensor interfacing, deep learning model testing, and data collection for gait-analysis experiments.

PROJECTS

Embedded Fault-Tolerant Attitude Monitoring System

- Built a fault-tolerant attitude monitoring system using STM32F303 and BNO055 IMU with real-time orientation tracking and automated anomaly recovery.
- Engineered robust error detection algorithms for sensor freeze, signal spikes, and disconnection events with system state visualization using onboard LEDs.
- Logged telemetry data over UART and simulated aerospace-grade reliability through watchdog timers and redundant attitude cross-validation

• Serial Communication • Sensor Interfacing • Microcontroller Programming • IMU • STM32F303

Anomaly Detection for Hot Axle Monitoring

- Developed an end-to-end intelligent wearable and software platform for real-time gait assessment and therapy.
- Engineered a custom wearable device embedded with motion sensors to accurately capture positional data movements in real time.
- Integrated Kinect-based motion capturing with wearable sensor data to enhance spatial resolution and improve detection accuracy of gait abnormalities.
- Developed a virtual therapy module delivering tailored rehabilitation programs based on dynamic movement patterns and improvement tracking.

• GY-906 • Esp32c3 • I²C Communication • Kalman filter • ESP-IDF • BNO055

Gait Revive - Beyond the walk

- Developed a pipeline for real-time detection of Hot Axle condition in railway systems using sensor time-series data (temperature, humidity, vibration axes).
- Applied a Kalman Filter to extract true system state from noisy sensor data, enabling reliable fusion and filtering of vibration and temperature data.
- Implemented unsupervised learning (DBSCAN, Hierarchical Clustering) to classify system states into Normal, Warning, and Critical without labeled data.

• BNO055 • Esp32c3 • Arduino IDE • Xbox - one • MATLAB
• Firebase • Web Dev • C++ • Deep Learning

KEY ACCOMPLISHMENTS AND CERTIFICATIONS

- Secured a ₹1 lakh **UG fellowship grant** for the **"Gait Revive"** project under Drishti CPS, **IIT Indore**, recognized in the BUILD program by **IIT Hyderabad's iTIC Incubator**, winning ₹1 Lakh and crucial support. Achieved second place and a ₹1.5 Lakh cash prize in the Take Off competition by UPES and Runway Incubator for "Gait Revive." Presented the project at TSIC Summit 3.0 and introduced it at the International Conference on IPR and SME, sponsored by the Ministry of MSME.
- Won the FIRST PRIZE for the idea " e-Setch Bin" in Social Innovation Hackathon organized by Telangana State Innovation Cell (TSIC) in T- Innovation Mahotsavam
- A **patent published for the projects "Gait Revive" and "Sight Scribe"** marking a significant step towards protecting and securing intellectual property rights for this innovative initiative.
- Presented a research paper titled "Smart Vest for Women Undergoing Menopause organized by Springer.
- Attended a 2 day workshop organised by Vertulonix on IoT devices.
- Participated in a mentorship program organized by **Qualcomm**.
- Certified in **MATLAB Onramp**: Proficient in MATLAB programming, data analysis, and algorithm development.